

■ ProductControllerTest.java – Tanglish Explanation

■ File Overview

Ithu `ProductControllerTest.java` file — idhu unit testing file for Product Controller. Ithu use panni namma API endpoints (GET, POST, PUT, DELETE) ellam correct-a work panratha check panrom.

■ Annotations Explanation

- `@SpringBootTest`: Entire Spring context load aagum testing time la.
- `@AutoConfigureMockMvc`: MockMvc use panna API test panna easy.
- `@Transactional`: Test mudinju data DB la save aagama rollback aagum.

■ Dependencies Used

1■■■ `MockMvc` – Simulate API requests. 2■■■ `ObjectMapper` – Java object ↔ JSON convert. 3■■■ `ProductRepository` – DB la data check panna use pannuranga.

■ Test Cases Breakdown

1. `**testGetAllProducts()**` → `/api/products` endpoint test. Expect: 200 OK + JSON content. 2. `**testGetProductById()**` → DB la first product eduthu GET pannum. Expect: correct product details. 3. `**testAddProduct()**` → POST test for adding new product. Expect: 201 Created + JSON verify. 4. `**testUpdateProduct()**` → PUT test for updating existing product quantity. Expect: 200 OK. 5. `**testDeleteProduct()**` → DELETE test to remove product. Expect: 204 No Content.

■ Important Note

Every test runs independently. `@Transactional` naala, data DB la save aagama rollback aagum. So real data affect aagala — safe testing environment.

■ Summary

Ithu one of the most important testing file. Product CRUD functionality correct-a iruka nu MockMvc moolama verify pannum. Automated testing nala future bug chance kammi aagum ■.